

Little Blue Dots Can't Reionize Helium Alone

Super-Eddington accretion in JWST's Little Blue Dots would end helium reionization too early.

arXiv:2608.28745 — Super-Eddington Little Blue Dots May Reionize Helium Too Early

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Background

JWST found many faint active galactic nuclei at cosmic dawn, the Little Blue Dots. Some models invoke super-Eddington accretion to explain their extreme ultraviolet emission.

Did these super-Eddington black holes reionize helium too early?

What this paper is selling

The paper tests whether super-Eddington Little Blue Dots can drive helium reionization. It finds that only a small fraction can be super-Eddington without violating timing constraints.

Key results to remember

- Only $\lesssim 10\%$ of Little Blue Dots can be super-Eddington.
- Majority super-Eddington requires escape fractions below 1.5%.
- Constraints depend on uncertain extreme-ultraviolet spectra.

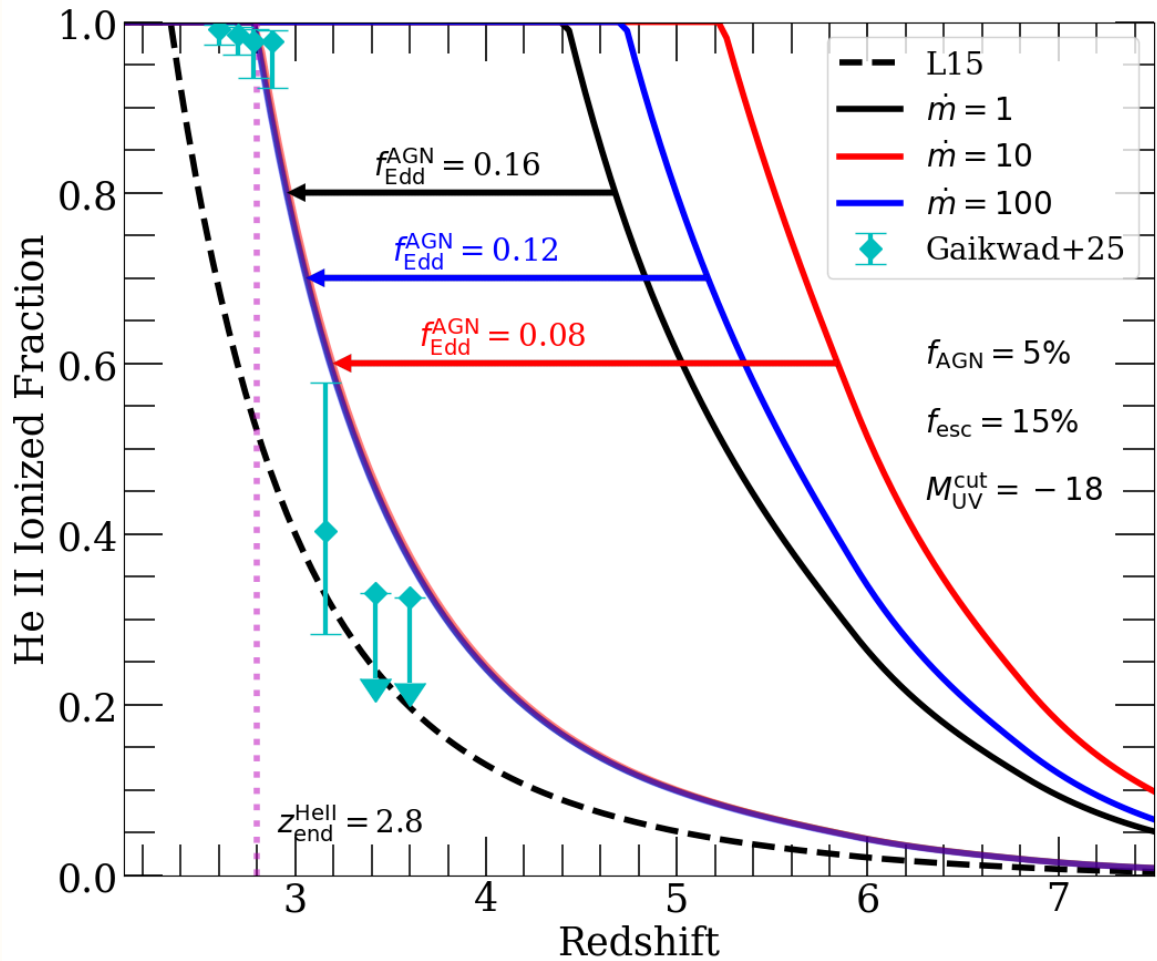


Figure 3 | Helium reionization histories show that super-Eddington models complete reionization too early unless the AGN fraction is small.

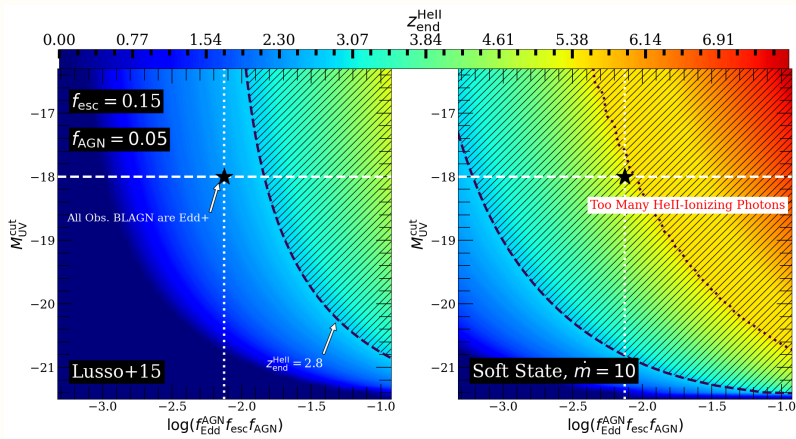


Figure 4 | Excluded parameter space reveals that high super-Eddington fractions and escape fractions are disfavored by helium reionization timing.